

13.

Ans: C

Condition for maxima: path difference = $n\lambda$ (and $\lambda = \frac{v}{f}$)

At one maxima: $2.9 = n \frac{v}{800}$ (1)

At next maxima: $2.9 = (n + 1) \frac{v}{960}$ (2)

Solve simultaneously gives $v = 460 \text{ m s}^{-1}$

Note: speed of sound is different in different medium, in air it is 340 m s^{-1} , in water it is 1500 m s^{-1} , in glass 4540 m s^{-1} , in hydrogen 1320 m s^{-1} , in neon it is 460 m s^{-1} .

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Ans: B